

1.6 Trace, determinants and partitioned matrices

1.6.1 Trace

The **trace** of a square matrix is the sum of its diagonal entries, $\text{tr}(\mathbf{A}) = \sum_i a_{ii}$. It is linear, $\text{tr}(a\mathbf{A} + b\mathbf{B}) = a\text{tr}(\mathbf{A}) + b\text{tr}(\mathbf{B})$, and $\text{tr}(\mathbf{A}^\top) = \text{tr}(\mathbf{A})$.

Theorem 1.6.1 — Cyclic property

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times m$, then $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.

Proof. $\text{tr}(\mathbf{AB}) = \sum_{i=1}^m \sum_{k=1}^n a_{ik} b_{ki} = \sum_{k=1}^n \sum_{i=1}^m b_{ki} a_{ik} = \text{tr}(\mathbf{BA})$. ■

The two products need not have the same size. Applied repeatedly, the theorem allows cyclic permutations such as $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB}) = \text{tr}(\mathbf{BCA})$, but not arbitrary reorderings. Three consequences occur again and again:

$$\text{tr}(\mathbf{A}^\top \mathbf{A}) = \sum_{i,j} a_{ij}^2 = \|\mathbf{A}\|_F^2, \quad \mathbf{x}^\top \mathbf{A} \mathbf{x} = \text{tr}(\mathbf{A} \mathbf{x} \mathbf{x}^\top), \quad \text{tr}(\mathbf{P}^{-1} \mathbf{A} \mathbf{P}) = \text{tr}(\mathbf{A}). \quad (1.6.1)$$

The middle one turns a scalar quadratic form into the trace of a matrix that is linear in $\mathbf{x} \mathbf{x}^\top$. Taking expectations then gives the mean of a random quadratic form ([Chapter 2](#)).

1.6.2 Determinants

The **determinant** of an $n \times n$ matrix is

$$\det \mathbf{A} = \sum_{\pi} \text{sgn}(\pi) a_{1\pi(1)} a_{2\pi(2)} \cdots a_{n\pi(n)},$$

the sum over all permutations π of $\{1, \dots, n\}$, where $\text{sgn}(\pi) = \pm 1$ according to whether π is a product of an even or odd number of transpositions. We collect the standard properties. Parts (a)–(d) are proved in any linear algebra text (Harville 1997, chapter 13), and the others follow from them.

Proposition 1.6.2 — Determinants

Let $\mathbf{A}, \mathbf{B}$ be $n \times n$.

- (a) $\det \mathbf{A}^\top = \det \mathbf{A}$.
- (b) $\det \mathbf{A}$ is a linear function of each column separately. It changes sign when two columns are interchanged, and is unchanged when a multiple of one column is added to another. The same holds for rows.

- (c) A triangular matrix has determinant equal to the product of its diagonal entries.
- (d) $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$.
- (e) $\mathbf{A}$ is nonsingular iff $\det \mathbf{A} \neq 0$, and then $\det \mathbf{A}^{-1} = 1/\det \mathbf{A}$.
- (f) $\det(c\mathbf{A}) = c^n \det \mathbf{A}$, and $\det \mathbf{Q} = \pm 1$ for orthogonal $\mathbf{Q}$.

Proof (Proof of (e) and (f)). If $\mathbf{A}$ is nonsingular, $1 = \det \mathbf{I} = \det \mathbf{A} \det \mathbf{A}^{-1}$ by (d). If $\mathbf{A}$ is singular, some column is a combination of the others. Subtracting that combination leaves a zero column without changing the determinant, by (b), and the determinant of a matrix with a zero column is zero, since every term of the sum contains a factor from that column. For (f), apply (b) to each of the n columns, and use (a) and (d) on $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$.

■

1.6.3 Schur complements and block determinants

Partitioned matrices arise whenever the regressors split into two groups, the model matrix $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ giving

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} \mathbf{X}_1^\top \mathbf{X}_1 & \mathbf{X}_1^\top \mathbf{X}_2 \\ \mathbf{X}_2^\top \mathbf{X}_1 & \mathbf{X}_2^\top \mathbf{X}_2 \end{pmatrix},$$

and whenever a random vector splits into two blocks (Chapter 3). In what follows

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}, \quad (1.6.2)$$

with $\mathbf{A}_{11}$ of size $k \times k$ and $\mathbf{A}_{22}$ of size $(n - k) \times (n - k)$.

Definition 1.6.1 — Schur complement

If $\mathbf{A}_{11}$ is nonsingular, the **Schur complement** of $\mathbf{A}_{11}$ in $\mathbf{A}$ is $\mathbf{A}_{22 \cdot 1} = \mathbf{A}_{22} - \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$.
If $\mathbf{A}_{22}$ is nonsingular, the Schur complement of $\mathbf{A}_{22}$ is $\mathbf{A}_{11 \cdot 2} = \mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21}$.

Everything in this subsection comes from one identity, block Gaussian elimination. If $\mathbf{A}_{11}$ is nonsingular, multiplying out shows

$$\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22 \cdot 1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}. \quad (1.6.3)$$

The outer factors are block triangular with identity blocks on the diagonal, and are nonsingular with inverses obtained by changing the sign of the off-diagonal block.

Theorem 1.6.3 — Block determinants

Let $\mathbf{A}$ be partitioned as in Eq. 1.6.2.

- (a) If $\mathbf{A}_{12} = \mathbf{0}$ or $\mathbf{A}_{21} = \mathbf{0}$, then $\det \mathbf{A} = \det \mathbf{A}_{11} \det \mathbf{A}_{22}$.
- (b) If $\mathbf{A}_{11}$ is nonsingular, $\det \mathbf{A} = \det \mathbf{A}_{11} \det \mathbf{A}_{22 \cdot 1}$. If $\mathbf{A}_{22}$ is nonsingular, $\det \mathbf{A} = \det \mathbf{A}_{22} \det \mathbf{A}_{11 \cdot 2}$.
- (c) For $m \times n$ matrices $\mathbf{U}$ and $n \times m$ matrices $\mathbf{V}$, $\det(\mathbf{I}_m + \mathbf{UV}) = \det(\mathbf{I}_n + \mathbf{VU})$. If $\mathbf{B}$ is $m \times m$ and nonsingular, $\det(\mathbf{B} + \mathbf{UV}) = \det \mathbf{B} \det(\mathbf{I}_n + \mathbf{VB}^{-1}\mathbf{U})$.

Proof.

- (a) Suppose $\mathbf{A}_{12} = \mathbf{0}$ (otherwise transpose). In the permutation expansion, a term that is nonzero cannot use an entry of the zero block $\mathbf{A}_{12}$. So each of the first k rows takes its entry from the first k columns, and π maps $\{1, \dots, k\}$ onto itself. Such a permutation splits into a permutation of the first k indices and one of the rest. Its sign is the product of their signs, so the sum factors as $\det \mathbf{A}_{11} \det \mathbf{A}_{22}$.
- (b) Take determinants in Eq. 1.6.3, using (a) and Proposition 1.6.2(d). The outer factors have determinant 1. The second statement follows in the same way from the elimination that starts with $\mathbf{A}_{22}$.
- (c) Apply (b) in both ways to $\begin{pmatrix} \mathbf{I}_m & -\mathbf{U} \\ \mathbf{V} & \mathbf{I}_n \end{pmatrix}$, whose Schur complements are $\mathbf{I}_n + \mathbf{VU}$ and $\mathbf{I}_m + \mathbf{UV}$. For the second statement, write $\mathbf{B} + \mathbf{UV} = \mathbf{B}(\mathbf{I}_m + \mathbf{B}^{-1}\mathbf{UV})$ and apply the first.

■

Part (c) with $n = 1$ is the **matrix determinant lemma** $\det(\mathbf{B} + \mathbf{uv}^\top) = \det \mathbf{B} (1 + \mathbf{v}^\top \mathbf{B}^{-1} \mathbf{u})$.

1.6.4 Inverses of partitioned matrices

Theorem 1.6.4 — Partitioned inverse

Let $\mathbf{A}$ be partitioned as in Eq. 1.6.2 with $\mathbf{A}_{11}$ nonsingular. Then $\mathbf{A}$ is nonsingular iff $\mathbf{A}_{22 \cdot 1}$ is, and in that case

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \\ -\mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{A}_{22 \cdot 1}^{-1} \end{pmatrix}. \quad (1.6.4)$$

If $\mathbf{A}_{22}$ is also nonsingular, the leading block of $\mathbf{A}^{-1}$ equals $\mathbf{A}_{11 \cdot 2}^{-1}$:

$$(\mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21})^{-1} = \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1}. \quad (1.6.5)$$

Proof. The nonsingularity statement is Theorem 1.6.3(b). Inverting the three factors of Eq. 1.6.3 and multiplying them in reverse order gives

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{I} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_{11}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22 \cdot 1}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix},$$

which multiplies out to Eq. 1.6.4. Eliminating with $\mathbf{A}_{22}$ first gives a second expression for $\mathbf{A}^{-1}$, whose leading block is $\mathbf{A}_{11 \cdot 2}^{-1}$. The inverse is unique, so the two leading blocks agree.

For a symmetric positive definite $\mathbf{X}^\top \mathbf{X}$ (Section 1.7), the theorem says that the block of $(\mathbf{X}^\top \mathbf{X})^{-1}$ belonging to $\mathbf{X}_1$ is $(\mathbf{X}_1^\top \mathbf{X}_1 - \mathbf{X}_1^\top \mathbf{X}_2 (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top \mathbf{X}_1)^{-1}$. Chapter 6 derives the same formula geometrically, from the Frisch–Waugh–Lovell theorem.

Identity Eq. 1.6.5 is a disguised form of a result that deserves its own name. Rename $\mathbf{A}_{11} = \mathbf{B}$, $\mathbf{A}_{12} = \mathbf{U}$, $\mathbf{A}_{21} = \mathbf{V}^\top$ and $\mathbf{A}_{22} = -\mathbf{C}^{-1}$.

Theorem 1.6.5 — Sherman–Morrison–Woodbury

Let $\mathbf{B}$ ($n \times n$) and $\mathbf{C}$ ($k \times k$) be nonsingular, and let $\mathbf{U}, \mathbf{V}$ be $n \times k$. If $\mathbf{C}^{-1} + \mathbf{V}^\top \mathbf{B}^{-1} \mathbf{U}$ is nonsingular, then so is $\mathbf{B} + \mathbf{UCV}^\top$, and

$$(\mathbf{B} + \mathbf{UCV}^\top)^{-1} = \mathbf{B}^{-1} - \mathbf{B}^{-1} \mathbf{U} (\mathbf{C}^{-1} + \mathbf{V}^\top \mathbf{B}^{-1} \mathbf{U})^{-1} \mathbf{V}^\top \mathbf{B}^{-1}. \quad (1.6.6)$$

In particular, for vectors $\mathbf{u}, \mathbf{v}$ with $1 + \mathbf{v}^\top \mathbf{B}^{-1} \mathbf{u} \neq 0$,

$$(\mathbf{B} + \mathbf{uv}^\top)^{-1} = \mathbf{B}^{-1} - \frac{\mathbf{B}^{-1} \mathbf{uv}^\top \mathbf{B}^{-1}}{1 + \mathbf{v}^\top \mathbf{B}^{-1} \mathbf{u}}. \quad (1.6.7)$$

Proof. Apply Theorem 1.6.4 to $\mathbf{N} = \begin{pmatrix} \mathbf{B} & \mathbf{U} \\ \mathbf{V}^\top & -\mathbf{C}^{-1} \end{pmatrix}$. Both diagonal blocks are nonsingular. The Schur complement of $\mathbf{B}$ is $-(\mathbf{C}^{-1} + \mathbf{V}^\top \mathbf{B}^{-1} \mathbf{U})$, which is nonsingular by hypothesis, so $\mathbf{N}$ is nonsingular. Then the Schur complement of the lower block, $\mathbf{B} + \mathbf{UCV}^\top$, is nonsingular by the same theorem with the roles of the blocks exchanged, and Eq. 1.6.5 becomes Eq. 1.6.6. For Eq. 1.6.7 take $k = 1$ and $\mathbf{C} = 1$.

Idea. Every result in this section comes from one block elimination, Eq. 1.6.3: a partitioned matrix is a unit triangular matrix times $\text{diag}(\mathbf{A}_{11}, \mathbf{A}_{22.1})$ times another. Taking determinants gives Theorem 1.6.3, inverting the factors gives Theorem 1.6.4, and eliminating in the other order and comparing gives the Woodbury identity. When a formula about a block is forgotten, redo the elimination.

The identity replaces the inversion of an $n \times n$ matrix by that of a $k \times k$ matrix. That matters when $\mathbf{B}^{-1}$ is already known or cheap, as in the following two examples.

Example. (Deleting an observation)

Removing the i th row $\mathbf{x}_i^\top$ from a model matrix changes $\mathbf{X}^\top \mathbf{X}$ into $\mathbf{X}^\top \mathbf{X} - \mathbf{x}_i \mathbf{x}_i^\top$. With $\mathbf{u} = -\mathbf{x}_i$, $\mathbf{v} = \mathbf{x}_i$ and $h_i = \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i$, Eq. 1.6.7 gives

$$(\mathbf{X}^\top \mathbf{X} - \mathbf{x}_i \mathbf{x}_i^\top)^{-1} = (\mathbf{X}^\top \mathbf{X})^{-1} + \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1}}{1 - h_i},$$

valid whenever $h_i < 1$, and the determinant lemma gives $\det(\mathbf{X}^\top \mathbf{X} - \mathbf{x}_i \mathbf{x}_i^\top) = (1 - h_i) \det(\mathbf{X}^\top \mathbf{X})$. The quantity h_i is the leverage of observation i (Chapter 6), and these

formulas are what make the deletion diagnostics of [Chapter 20](#) cheap (Theorem 20.3.1). For Brownlee's stack-loss data ($n = 21$, $p = 4$), the listing updates the inverse for each of the 21 deletions. The largest relative discrepancy from direct inversion is 1×10^{-13} . Observation 17 has the largest leverage, 0.412, and removing it multiplies $\det(\mathbf{X}^\top \mathbf{X})$ by 0.588.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.stackloss.load_pandas().data
X = np.column_stack([np.ones(len(data)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"
]]])
n, p = X.shape

A_inv = np.linalg.inv(X.T @ X)
h = np.einsum("ij,jk,ik->i", X, A_inv, X)           # leverages  $x_i' (X'X)^{-1} x_i$ 
x_i

worst = 0.0
for i in range(n):
    x = X[i]
    # Sherman-Morrison with  $u = -x$ ,  $v = x$ :  $(X'X - x x')^{-1}$ 
    B_update = A_inv + np.outer(A_inv @ x, x @ A_inv) / (1 - h[i])
    X_del = np.delete(X, i, axis=0)
    B_direct = np.linalg.inv(X_del.T @ X_del)
    worst = max(worst, np.abs(B_update - B_direct).max() / np.abs(B_direct).max())
print(f"largest relative discrepancy over all {n} deletions: {worst:.1e}")
```

[Run this code online](#)

Example. (Diagonal plus low rank)

In factor models and in the linear mixed models of [Chapter 32](#), covariance matrices have the form $\Sigma = \mathbf{D} + \mathbf{L}\mathbf{L}^\top$ with $\mathbf{D}$ diagonal and positive and $\mathbf{L}$ of size $N \times k$, $k \ll N$. By [Eq. 1.6.6](#) and [Theorem 1.6.3\(c\)](#),

$$\Sigma^{-1} = \mathbf{D}^{-1} - \mathbf{D}^{-1}\mathbf{L}(\mathbf{I}_k + \mathbf{L}^\top \mathbf{D}^{-1}\mathbf{L})^{-1}\mathbf{L}^\top \mathbf{D}^{-1}, \quad \det \Sigma = \det \mathbf{D} \det(\mathbf{I}_k + \mathbf{L}^\top \mathbf{D}^{-1}\mathbf{L}).$$

Each needs only a $k \times k$ factorization. The script checks this with $N = 1500$ and $k = 4$. The solution of $\Sigma \mathbf{w} = \mathbf{v}$ agrees with a direct solve to a relative error of 8×10^{-14} .

Example. (Equicorrelation)

The matrix $\mathbf{E} = (1 - \rho)\mathbf{I}_n + \rho\mathbf{J}_n$ has unit diagonal and every off-diagonal entry equal to ρ . It is the correlation matrix of exchangeable observations. Taking $\mathbf{B} = (1 - \rho)\mathbf{I}$ and $\mathbf{u} = \rho\mathbf{1}$, $\mathbf{v} = \mathbf{1}$, for $\rho \neq 1$ and $1 + (n - 1)\rho \neq 0$,

$$\mathbf{E}^{-1} = \frac{1}{1 - \rho} \left(\mathbf{I}_n - \frac{\rho}{1 + (n - 1)\rho} \mathbf{J}_n \right), \quad \det \mathbf{E} = (1 - \rho)^{n-1} (1 + (n - 1)\rho).$$

1.6.5 Exercises

1.6.5.1 A. Check your understanding

Exercise A1.

Show that $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{ACB})$ when $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are symmetric, and find 2×2 matrices for which the two traces differ.

Solution. For symmetric matrices, $\text{tr}(\mathbf{ABC}) = \text{tr}((\mathbf{ABC})^\top) = \text{tr}(\mathbf{CBA}) = \text{tr}(\mathbf{ACB})$, using transposition and then the cyclic property. For a counterexample take $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Then $\mathbf{ABC} = \mathbf{C}$ has trace 1, while $\mathbf{AC} = \mathbf{0}$ gives $\text{tr}(\mathbf{ACB}) = 0$.

Exercise A2.

Let $\mathbf{A}$ be nonsingular, $\mathbf{a}$ a vector and c a scalar with $s = c - \mathbf{a}^\top \mathbf{A}^{-1} \mathbf{a} \neq 0$. Write down the inverse and the determinant of $\begin{pmatrix} \mathbf{A} & \mathbf{a} \\ \mathbf{a}^\top & c \end{pmatrix}$. Use it to update $(\mathbf{X}^\top \mathbf{X})^{-1}$ when a column is added to $\mathbf{X}$.

1.6.5.2 B. Practice

Exercise B1.

Let $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, $e_i = y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$ and $h_i = \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i < 1$. Show that the least squares coefficients computed without observation i are

$$\hat{\boldsymbol{\beta}}_{(i)} = \hat{\boldsymbol{\beta}} - \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i e_i}{1 - h_i}.$$

Solution. Write $\mathbf{K} = (\mathbf{X}^\top \mathbf{X})^{-1}$ and $\mathbf{x} = \mathbf{x}_i$. Deleting row i turns $\mathbf{X}^\top \mathbf{y}$ into $\mathbf{X}^\top \mathbf{y} - \mathbf{x} y_i$, so by [Example \(Deleting an observation\)](#)

$$\hat{\boldsymbol{\beta}}_{(i)} = \left(\mathbf{K} + \frac{\mathbf{K} \mathbf{x} \mathbf{x}^\top \mathbf{K}}{1 - h_i} \right) (\mathbf{X}^\top \mathbf{y} - \mathbf{x} y_i) = \hat{\boldsymbol{\beta}} - \mathbf{K} \mathbf{x} y_i + \frac{\mathbf{K} \mathbf{x} (\mathbf{x}^\top \hat{\boldsymbol{\beta}} - h_i y_i)}{1 - h_i} = \hat{\boldsymbol{\beta}} - \mathbf{K} \mathbf{x} \frac{y_i - \mathbf{x}^\top \hat{\boldsymbol{\beta}}}{1 - h_i},$$

since $(1 - h_i) y_i - \mathbf{x}^\top \hat{\boldsymbol{\beta}} + h_i y_i = y_i - \mathbf{x}^\top \hat{\boldsymbol{\beta}}$.

Exercise B2.

For an $n \times p$ matrix $\mathbf{X}$ and $\lambda > 0$, show that $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}_n)^{-1}$. Explain why the right-hand side is preferable when p is much larger than n , and derive the identity a second way from [Theorem 1.6.5](#).

Solution. Both $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ and $\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}$ are positive definite. The identity $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p) \mathbf{X}^\top = \mathbf{X}^\top \mathbf{X} \mathbf{X}^\top + \lambda \mathbf{X}^\top = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}_n)$ gives the result after multiplying by the two inverses. When $p \gg n$, the right-hand side inverts an $n \times n$ matrix instead of a $p \times p$ one. For the second derivation, apply [Theorem 1.6.5](#) with $\mathbf{B} = \lambda \mathbf{I}_p$, $\mathbf{U} = \mathbf{X}^\top$, $\mathbf{C} = \mathbf{I}_n$ and $\mathbf{V} = \mathbf{X}^\top$: $(\lambda \mathbf{I}_p + \mathbf{X}^\top \mathbf{X})^{-1} = \lambda^{-1} \mathbf{I}_p - \lambda^{-2} \mathbf{X}^\top \mathbf{W}^{-1} \mathbf{X}$ with $\mathbf{W} = \mathbf{I}_n + \mathbf{X} \mathbf{X}^\top / \lambda$. Multiplying

on the right by $\mathbf{X}^\top$ gives $\lambda^{-1}\mathbf{X}^\top[\mathbf{I}_n - \mathbf{W}^{-1}(\mathbf{X}\mathbf{X}^\top/\lambda)] = \lambda^{-1}\mathbf{X}^\top\mathbf{W}^{-1}[\mathbf{W} - \mathbf{X}\mathbf{X}^\top/\lambda] = \lambda^{-1}\mathbf{X}^\top\mathbf{W}^{-1} = \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top + \lambda\mathbf{I}_n)^{-1}$.

1.6.5.3 C. Going deeper

Exercise C1.

Let $\mathbf{A}$ be positive definite of order n . Write $\mathbf{A}_{-j}$ for $\mathbf{A}$ with its j th row and column deleted and $\mathbf{a}_j$ for its j th column with the j th entry removed. Show that

$$[\mathbf{A}^{-1}]_{jj} = \frac{1}{a_{jj} - \mathbf{a}_j^\top \mathbf{A}_{-j}^{-1} \mathbf{a}_j} \geq \frac{1}{a_{jj}},$$

with equality iff $\mathbf{a}_j = \mathbf{0}$. Now let $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$, where the columns of $\mathbf{X}$ have been centred and scaled to unit length, and put $R_j^2 = \mathbf{a}_j^\top \mathbf{A}_{-j}^{-1} \mathbf{a}_j$, the squared multiple correlation of the j th regressor with the others. Show $0 \leq R_j^2 < 1$ and $[\mathbf{A}^{-1}]_{jj} = 1/(1 - R_j^2)$. This is the *variance inflation factor* of [Section 26.2](#): the price in variance paid for regressors that the others can nearly reproduce.

Solution. Let $\mathbf{P}$ be the permutation matrix that moves index j to the last position. Then $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \begin{pmatrix} \mathbf{A}_{-j} & \mathbf{a}_j \\ \mathbf{a}_j^\top & a_{jj} \end{pmatrix}$ is positive definite, and its inverse is $\mathbf{P}^\top \mathbf{A}^{-1} \mathbf{P}$, whose last diagonal entry is $[\mathbf{A}^{-1}]_{jj}$. By [Theorem 1.6.4](#) the trailing block of that inverse is the reciprocal of the Schur complement $a_{jj} - \mathbf{a}_j^\top \mathbf{A}_{-j}^{-1} \mathbf{a}_j$, which is positive by [Proposition 1.7.5\(e\)](#). Since $\mathbf{A}_{-j}^{-1}$ is positive definite, the subtracted term is nonnegative, and zero only if $\mathbf{a}_j = \mathbf{0}$; this gives the inequality.

With unit-length columns, $a_{jj} = \mathbf{x}_j^\top \mathbf{x}_j = 1$, so the Schur complement is $1 - R_j^2$, which is positive, while $R_j^2 \geq 0$ again because $\mathbf{A}_{-j}^{-1}$ is positive definite. Hence $[\mathbf{A}^{-1}]_{jj} = 1/(1 - R_j^2)$.

Exercise C2.

Let $\mathbf{A}$ and $\mathbf{B}$ be $k \times k$ and let $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{A} \end{pmatrix}$. Show that $\mathbf{P} = 2^{-1/2} \begin{pmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{I} & -\mathbf{I} \end{pmatrix}$ is symmetric and orthogonal and that $\mathbf{P} \mathbf{M} \mathbf{P} = \text{diag}(\mathbf{A} + \mathbf{B}, \mathbf{A} - \mathbf{B})$. Deduce that $\det \mathbf{M} = \det(\mathbf{A} + \mathbf{B}) \det(\mathbf{A} - \mathbf{B})$, that $\mathbf{M}$ is nonsingular iff both $\mathbf{A} + \mathbf{B}$ and $\mathbf{A} - \mathbf{B}$ are, and that then

$$\mathbf{M}^{-1} = \frac{1}{2} \begin{pmatrix} \mathbf{S} + \mathbf{D} & \mathbf{S} - \mathbf{D} \\ \mathbf{S} - \mathbf{D} & \mathbf{S} + \mathbf{D} \end{pmatrix}, \quad \mathbf{S} = (\mathbf{A} + \mathbf{B})^{-1}, \quad \mathbf{D} = (\mathbf{A} - \mathbf{B})^{-1}.$$

If $\mathbf{M}$ is the covariance matrix of two exchangeable blocks of measurements, what does the change of coordinates $\mathbf{P}$ do to them?

Solution. $\mathbf{P}^\top = \mathbf{P}$ and $\mathbf{P}^\top \mathbf{P} = \frac{1}{2} \begin{pmatrix} 2\mathbf{I} & \mathbf{0} \\ \mathbf{0} & 2\mathbf{I} \end{pmatrix} = \mathbf{I}$. Multiplying out,

$$\mathbf{P} \mathbf{M} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{A} + \mathbf{B} & \mathbf{A} + \mathbf{B} \\ \mathbf{A} - \mathbf{B} & \mathbf{B} - \mathbf{A} \end{pmatrix}, \quad (\mathbf{P} \mathbf{M}) \mathbf{P} = \begin{pmatrix} \mathbf{A} + \mathbf{B} & \mathbf{0} \\ \mathbf{0} & \mathbf{A} - \mathbf{B} \end{pmatrix},$$

with no symmetry of $\mathbf{A}$ or $\mathbf{B}$ needed. Taking determinants and using [Proposition 1.6.2\(d\)](#), (f) and [Theorem 1.6.3\(a\)](#) gives $\det \mathbf{M} = \det(\mathbf{A} + \mathbf{B}) \det(\mathbf{A} - \mathbf{B})$, since $(\det \mathbf{P})^2 = 1$. As $\mathbf{M} = \mathbf{P} \text{diag}(\mathbf{A} + \mathbf{B}, \mathbf{A} - \mathbf{B}) \mathbf{P}$, the matrix $\mathbf{M}$ is nonsingular iff the two blocks are, and

then $\mathbf{M}^{-1} = \mathbf{P} \text{diag}(\mathbf{S}, \mathbf{D}) \mathbf{P}$, which multiplies out to the displayed matrix.

For a covariance matrix, $\mathbf{P}$ replaces the two blocks $\mathbf{y}_1, \mathbf{y}_2$ by the scaled sum and difference $(\mathbf{y}_1 + \mathbf{y}_2)/\sqrt{2}$ and $(\mathbf{y}_1 - \mathbf{y}_2)/\sqrt{2}$. These are uncorrelated, with covariance matrices $\mathbf{A} + \mathbf{B}$ and $\mathbf{A} - \mathbf{B}$: exchangeable blocks separate into a between-pair part and a within-pair part, which is the algebra behind paired comparisons and split-plot analyses.

Exercise C3.

Let $\mathbf{A}$ be symmetric, partitioned as in [Eq. 1.6.2](#), with $\mathbf{A}_{11}$ positive definite. Using [Eq. 1.6.3](#), show that $\mathbf{A}$ is nonnegative definite iff the Schur complement $\mathbf{A}_{22.1}$ is, and positive definite iff $\mathbf{A}_{22.1}$ is. Applying this to $[\mathbf{C}, \mathbf{B}]^\top [\mathbf{C}, \mathbf{B}]$, deduce the matrix inequality

$$\mathbf{B}^\top \mathbf{C} (\mathbf{C}^\top \mathbf{C})^{-1} \mathbf{C}^\top \mathbf{B} \preceq \mathbf{B}^\top \mathbf{B}$$

for any $\mathbf{C}$ of full column rank and any $\mathbf{B}$ with the same number of rows, and hence that the leverage $h_i = \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i$ of [Example \(Deleting an observation\)](#) satisfies $0 \leq h_i \leq 1$. (Nonnegative definiteness is defined in [Section 1.7](#).)

Solution. For symmetric $\mathbf{A}$ we have $\mathbf{A}_{21} = \mathbf{A}_{12}^\top$, so the two outer factors of [Eq. 1.6.3](#) are transposes of each other: $\mathbf{A} = \mathbf{T}^\top \text{diag}(\mathbf{A}_{11}, \mathbf{A}_{22.1}) \mathbf{T}$ with $\mathbf{T} = \begin{pmatrix} \mathbf{I} & \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$ nonsingular. Hence $\mathbf{x}^\top \mathbf{A} \mathbf{x} = (\mathbf{T} \mathbf{x})^\top \text{diag}(\mathbf{A}_{11}, \mathbf{A}_{22.1}) (\mathbf{T} \mathbf{x})$, and as $\mathbf{x}$ runs over $\mathbb{R}^n$ so does $\mathbf{T} \mathbf{x}$. Writing $\mathbf{T} \mathbf{x} = (\mathbf{u}^\top, \mathbf{v}^\top)^\top$, the right-hand side is $\mathbf{u}^\top \mathbf{A}_{11} \mathbf{u} + \mathbf{v}^\top \mathbf{A}_{22.1} \mathbf{v}$. Since $\mathbf{A}_{11} \succ \mathbf{0}$, this is nonnegative for all $\mathbf{u}, \mathbf{v}$ iff $\mathbf{A}_{22.1} \succeq \mathbf{0}$, and positive for all $(\mathbf{u}, \mathbf{v}) \neq \mathbf{0}$ iff $\mathbf{A}_{22.1} \succ \mathbf{0}$.

Now take $\mathbf{A} = [\mathbf{C}, \mathbf{B}]^\top [\mathbf{C}, \mathbf{B}]$, which is nonnegative definite, with $\mathbf{A}_{11} = \mathbf{C}^\top \mathbf{C}$ positive definite by Proposition 1.2.3(c). Its Schur complement is $\mathbf{B}^\top \mathbf{B} - \mathbf{B}^\top \mathbf{C} (\mathbf{C}^\top \mathbf{C})^{-1} \mathbf{C}^\top \mathbf{B}$, which is therefore nonnegative definite. With $\mathbf{C} = \mathbf{X}$ and $\mathbf{B} = \mathbf{e}_i$ the inequality reads $\mathbf{e}_i^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{e}_i \leq \mathbf{e}_i^\top \mathbf{e}_i = 1$, that is, $h_i \leq 1$; and $h_i \geq 0$ because $(\mathbf{X}^\top \mathbf{X})^{-1}$ is positive definite. A leverage is a number between zero and one, and [Example \(Deleting an observation\)](#) shows that the deletion formulas break down exactly at $h_i = 1$.